Process & Decision Documentation

Project/Assignment Decisions

This sidequest was primarily GenAI, but I had to make some adjustments myself. I got AI to generate the code for me and explain how to toggle moon gravity.

```
  // Created by GenAI      You, now • Uncommitted changes
  if (debugMode) {
    camera.off();

    // Background panel
    fill(0, 150);
    noStroke();
    rect(5, 5, 140, 60);

    // Text
    fill(255);
    textSize(10);
    textAlign(LEFT, TOP);

    text("DEBUG MENU", 10, 10);
    text("FPS: " + Math.round(frameRate()), 10, 22);
    text("Gravity: " + (moonGravity ? "MOON" : "NORMAL"), 10, 34);
    text("Press G: Toggle Gravity", 10, 46);

    camera.on();
  }
}
```

```
world.gravity.y = NORMAL_GRAVITY; // created by GenAI
    You, 23 minutes ago • initial commit
```

Annotation: code generated by GenAI at the bottom of “draw”

```
// --- DEBUG TOGGLE ---
if (key === "F3") {
  debugMode = !debugMode;
}
```

```
// edited by GenAI
function keyPressed() {
  startMusicIfNeeded();

  // --- DEBUG TOGGLE ---
  if (key === "[") {
    debugMode = !debugMode;
  }

  // --- MOON GRAVITY TOGGLE ---
  if (key === "g" || key === "G") {
    moonGravity = !moonGravity;
    world.gravity.y = moonGravity ? MOON_GRAVITY : NORMAL_GRAVITY;
  }
}      You, 22 minutes ago • initial commit …
```

Annotation: Changed debug key code from F3 to [

```
// --- DEBUG --- created by GenAI
let debugMode = false;
let moonGravity = false;

const NORMAL_GRAVITY = 10;
const MOON_GRAVITY = 1.6;
```

```
const NORMAL_GRAVITY = 10;
const MOON_GRAVITY = 4;
```

Annotation: Updated the moon gravity to be stronger. I figured that the closer to normal gravity value, the stronger it would be.

GenAI Documentation

Date Used: Friday, March 20, 2026

Tool Disclosure: ChatGPT v 5.2

Purpose of Use: I used GenAI to create the debug screen, add a moon gravity toggle, and add an FPS counter into the debug screen.

Summary of Interaction: I pasted in the sketch.js code so that the GenAI would know the code that already existed. From there, I told it to add a debug screen for the moon gravity toggle and an extra feature. It recommended a few to me, and I chose to add an FPS counter.

Human Decision Point(s): I chose which extra feature I wanted, and decided how strong I wanted the “moon gravity” value to be.

I had to change the key bind to turn the debug screen on because it was using a key that I don’t have on my computer.

Integrity & Verification Note: I made notes of where I added in the GenAI curated code, and I made sure not to delete any existing code, in case the new code didn’t work. The code worked as expected, so I didn’t need to make any changes other than the toggle key.

Scope of GenAI Use: GenAI did most of the work for this Sidequest. I still went over the code it provided to make sure it would actually be compatible (like the key change), but GenAI did most of the work.

Limitations or Misfires: The tool generated the code properly, but it had made the toggle key for the debug screen the F3 key, which I don’t have on a mac. ChatGPT had

no way of knowing that I was on a Mac without me telling it, which is a limitation that I hadn't thought of before that issue arose.

Summary of Process (Human + Tool)

The process is documented above, but this is the summary.

1. I fed the existing code and the instructions into the tool.
2. The GenAI gave me code and told me where in the existing code to put it.
3. I ran the new code to see if there were any errors. I realized that I didn't know how to toggle the gravity or the debug screen so I had to go back and see which keys triggered it in the code.
4. I updated the toggle key for the debug screen to be the "[" key, and I changed the moon gravity to be a little bit stronger so that you could still see the character on screen when you jump.

Decision Points & Trade-offs

1. I had to change the debug screen toggle key from "F3" to "["
2. I changed the moon's gravity to be stronger so the character would still be visible when jumping.

Verification & Judgement

I always begin with small changes to the code if I need something to be tweaked. With the key change, I asked the GenAI to change the key, and all it did was rewrite the code with the new key. I saw that nothing else changed, so I simply switched the value in the keyPressed function. I also tweaked the number of the moon's gravity to be higher (closer to normal gravity) and ran the code, seeing that it worked.

Limitations, Dead Ends, or Open Questions

As noted previously, the only limitation was that the GenAI didn't recognize my computer software. This isn't a big problem, as I just had to change the key value, but it is a limitation that I had not taken into consideration previously.

Appendix

Me = purple

GenAI = Green

I'm going to paste some base code that I've been given. I want you to add a debug screen that allows you to toggle moon gravity on/off, and at least one other potential feature you might need in developing a game.

Perfect—paste your base code and I'll wire the debug screen directly into it.

Before you send it, quick note on what I'll add so you know what to expect:

🔧 Planned Debug Features

- **Toggle Moon Gravity (on/off)**
  → Switch between normal gravity and ~1/6 gravity (like the Moon)

- **Second Debug Feature (useful for dev)**
  I'll include one of these depending on your setup:

  - **God Mode / No Damage toggle**
  - **Show Hitboxes toggle**
  - **FPS counter**
  - **Free camera / noclip**

- (If you have a preference, tell me before I start)

- **Debug UI**

  - Toggle with a key (e.g., F3 or `)
  - Simple overlay panel
  - Real-time state display (gravity mode, etc.)

Once you paste the code, I'll:

1. Integrate cleanly with your structure (no breaking your logic)
2. Keep it modular so you can expand later

3. Comment everything so you understand it for your assignment

Send it 👍

Include the "FPS counter" as the second debug feature. *Pasted sketch.js*

Nice, this is clean code—easy to extend 👍
I'll add a **debug system** that fits your current structure without breaking anything.

✅ What's Being Added

🎮 Toggleable Debug Menu

- Press **F3** → show/hide debug overlay

🌙 Moon Gravity Toggle

- Press `G` → switch between:
  - Normal gravity (`10`)
  - Moon gravity (`~1.6`)

📊 FPS Counter

- Always shown **when debug menu is open**

🧩 STEP 1: Add Debug Variables (top of file)

Put this near your other globals:

```
// --- DEBUG ---
let debugMode = false;
let moonGravity = false;

const NORMAL_GRAVITY = 10;
const MOON_GRAVITY = 1.6;
```

🧩 STEP 2: Update Gravity in `setup()`

Replace:

```
world.gravity.y = GRAVITY;
```

With:

```
world.gravity.y = NORMAL_GRAVITY;
```

🧩 STEP 3: Add Key Controls (inside `keyPressed()`)

Modify your existing function:

```
function keyPressed() {
 startMusicIfNeeded();

 // --- DEBUG TOGGLE ---
```

```
 if (key === "F3") {
   debugMode = !debugMode;
 }

 // --- MOON GRAVITY TOGGLE ---
 if (key === "g" || key === "G") {
   moonGravity = !moonGravity;
   world.gravity.y = moonGravity ? MOON_GRAVITY : NORMAL_GRAVITY;
 }
}
```

🧩 STEP 4: Add Debug Overlay (bottom of `draw()`)

Add this **at the VERY END of `draw()`**:

```
// --- DEBUG SCREEN ---
if (debugMode) {
 camera.off();

 // Background panel
 fill(0, 150);
 noStroke();
 rect(5, 5, 140, 60);
```

```
// Text
fill(255);
textSize(10);
textAlign(LEFT, TOP);

text("DEBUG MENU", 10, 10);
text("FPS: " + Math.round(frameRate()), 10, 22);
text("Gravity: " + (moonGravity ? "MOON" : "NORMAL"), 10, 34);
text("Press G: Toggle Gravity", 10, 46);

camera.on();
}
```

🧪 What You Can Now Do In-Game

Key	Action
F3	Toggle debug screen
G	Toggle moon gravity

(View FPS counter
)

💡 Why This Is Actually Useful (for your assignment)

This isn't just cosmetic—it shows **real dev practices**:

- Runtime testing tools
- Physics tuning (gravity switching)
- Performance monitoring (FPS)

🚀 Optional Upgrades (if you want extra marks)

I can help you add:

- Hitbox visibility toggle
- Player position display
- Slow motion mode
- Free camera

Just tell me 👍

I'm on a mac so can you change the debug screen toggle to the] key?

Good call—function keys on Mac can be annoying to deal with.

Let's switch the debug toggle to the] key.

🔁 Update Your `keyPressed()` Function

Replace your debug toggle section with this:

```
function keyPressed() {
 startMusicIfNeeded();

 // --- DEBUG TOGGLE ( ] key ) ---
 if (key === "]") {
   debugMode = !debugMode;
 }

 // --- MOON GRAVITY TOGGLE ---
 if (key === "g" || key === "G") {
   moonGravity = !moonGravity;
   world.gravity.y = moonGravity ? MOON_GRAVITY : NORMAL_GRAVITY;
 }
}
```

⚠️ Small Mac Note

On some keyboards:

- You might need **Shift +]** depending on layout
- If it doesn't trigger, use this more robust version:

```
if (key === "]" || keyCode === 221) {
 debugMode = !debugMode;
}
```

✅ Final Controls

Key	Action
`]`	Toggle debug menu
`G`	Toggle moon gravity
(View)	FPS counter

If you want, I can make the debug menu clickable (like real dev tools) or add sliders for gravity—that's a nice "wow factor" for assignments.